

SolaX Battery 3 Level Heating Solution



1. Introduction

In some high-latitude or high-altitude areas, the temperature (during winter) is quite low. The battery charging and discharging performance will be impacted significantly by the low temperature. SolaX developed a function of “Battery Heating” with 3 different levels to make your energy storage system able to work under extreme low temperature.

This function is disabled by default, so you need to **enable** this function if needed. After you enable this function, you also need to **set heating periods and heating level**.

2. Function logic

2.1 Low level

The cell temperature is below than **0°C**, it will be heated until **10°C**. Once the cell temperature is below **0°C** again, the battery system will be heated.

2.2 Medium level

The cell temperature is below than **10°C**, it will be heated until **15°C**. Once the cell temperature is below **10°C** again, the battery system will be heated.

2.3 High level

The cell temperature is below than **20°C**, it will be heated until **23°C (T30) or 26°C (HS25/36/50E/51)**. Once the cell temperature is below **20°C** again, the battery system will be heated.

Note:

- a. *Only if the cell temperature is higher than -10°C, the battery will be able to discharge, and only if cell temperature is higher than 0°C, the battery will be able to be charged.*
- b. *The heating power per module is 180W for per T30 battery, 120W for per HS25/HS36 battery, 328W for per HS50E battery and 346W for per HS51 battery.*
- c. *Please check the heating wires are correctly connected, or the heating function will not take effect.*
- d. *The heating control is based on the cell temperature, which the internal temperature rather than the environment temperature. Normally the cell temperature is higher than the environment.*

3. Heating during the heating periods

The heating system will take energy from **PV in priority**. If the PV power is not enough, it will take energy from **battery or grid**, which depends on the *BAT actual SOC*.

- If *BAT actual SOC* > 35%, it takes from **Battery**. After using the Battery for heating, when the SOC is below 25%, the system will switch to take from **Grid** for heating.
- If *BAT actual SOC* < 35%, it takes from **Grid**.

4. Heating during the non-heating periods

The battery will be heated **only if the PV has surplus energy**.

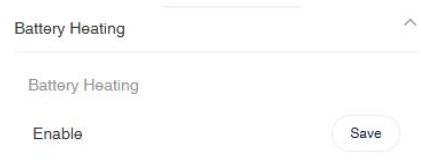
5. Configuration

Setting path:

Main → Settings → Advance Setting → Battery Heating

Step 1: Enable the 'Battery Heating' function.

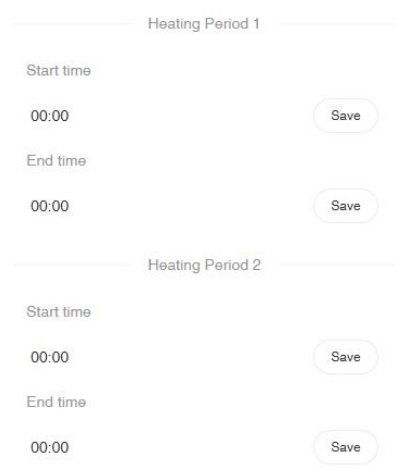
This function is only valid when the battery has the heating function.



The screenshot shows a mobile app interface for 'Battery Heating'. At the top, there's a title bar with 'Battery Heating' and an upward arrow. Below it, a card titled 'Battery Heating' contains a toggle switch labeled 'Enable' and a 'Save' button.

Step 2: Set the 'Heating Period' for the battery.

Two heating periods can be set. The setting range is 00:00 to 23:59.



The screenshot shows a mobile app interface for setting heating periods. It has two sections: 'Heating Period 1' and 'Heating Period 2'. Each section contains a 'Start time' field with a 'Save' button and an 'End time' field with a 'Save' button. Both time fields currently show '00:00'.

Step 3: Set different Heating Level.

There are 3 levels of temperature: Low, Medium and High. Once a battery is heated to the target temperature, heating will stop.

- Low: Target temperature: **10°C**. Battery charge/discharge current: **10A**.
- Medium: Target temperature: **15°C**. Battery charge/discharge current: **15A**.
- High: Target temperature: **23°C(T30) and 26°C(HS25/36/50E/51)**. Battery charge/discharge current: **30A**.



Note:

If the ambient temperature is extremely low, turning on battery heating will consume a significant amount of electrical energy.

Compatibility

Compatible Model List:

Inverter	Battery
X1-IES	T-BAT-SYS-HV 3.0
X3-IES (X3-IES-P)	T-BAT-SYS-HV S2.5/S3.6 (Only with X3-Ultra)
X1-Hybrid/Fit G4	T-BAT-SYS-HV S50E (Only with X3-IES)
X3-Hybrid/Fit G4	TSYS-HS51
X3-Ultra	
X1-Vast	

Note:

- Before using this function, ensure that the firmware of the battery and inverter has been upgraded to the latest version.*